RELATIVE M-PSEUDO-COHERENCE OF A COMPLEX IS AN FPPF CONDITION ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20699-20708 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20710-20721, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The fppf-base version of the chapter's pseudo-coherence descent family; short but complete, not a pointer.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas relative-pseudo-coherent-descends-fppf and relative-pseudo-coherent-post-compose (both previously collected inputs in this lane's queue), definitions of relative pseudo-coherence for morphisms locally of finite type.

Source reference. Original lemma 0CSN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20631–20637. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20683–20689. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be locally of finite type. Let $\{S_i \to S\}$ be an fppf covering of schemes. Denote $f_i : X_i \to S_i$ the base change of f and $g_i : X_i \to X$ the projection. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent relative to S if and only if each Lg_i^*E is m-pseudo-coherent relative to S_i.*

Proof. This follows formally from Lemmas 0CSN and 0CSP. Namely, if E is m-pseudo-coherent relative to S, then Lg_i^*E is m-pseudo-coherent relative to S (by the first lemma), hence Lg_i^*E is m-pseudo-coherent relative to S_i (by the second). Conversely, if Lg_i^*E is m-pseudo-coherent relative to S_i, then Lg_i^*E is m-pseudo-coherent relative to S (by the second lemma), hence E is m-pseudo-coherent relative to S (by the first lemma). □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.